UNEB BIOLOGY PRACTICAL GUIDE

Comprehensive Summary of Food Tests, Experimental Procedures & Control Variables

Tailored for Secondary Examination Revision & Practical Execution

1. Starch

BIOLOGICAL IMPORTANCE

Provides the primary and immediate source of energy required for cellular respiration, metabolic functions, and daily physical activity.

SIGNS, SYMPTOMS & DEFICIENCY

Deficiency: Marasmus (Severe Energy-Protein Malnutrition).
Symptoms: Extreme emaciation, muscle wasting, visible ribs, wrinkled skin, constant fatigue, and severe weight loss.

Summarised Practical Procedure

1. Measure ***2 cm^3*** of the test food solution (or place a solid piece of sample) into a clean, dry test tube.
2. Add ***2–3 drops*** of **Iodine solution** to the sample.
3. Gently shake the test tube and observe any color change.

Observation (Positive)	Observation (Negative)	Deduction / Conclusion
Color changes from brown / yellowish-brown to **blue-black**.	Solution remains **brown / yellowish-brown**.	Starch present (Positive) / Starch absent (Negative).

2. Proteins

BIOLOGICAL IMPORTANCE

Essential for growth, structural repair of damaged body tissues, synthesis of vital enzymes, hormones, hemoglobin, and protective antibodies.

SIGNS, SYMPTOMS & DEFICIENCY

Deficiency: Kwashiorkor.
Symptoms: Stunted growth, swollen abdomen and legs (edema), skin lesions/flaking, sparse reddish-brown hair, apathy, and weakened immunity.

Summarised Practical Procedure (Biuret Test)

1. Add ***2 cm^3*** of test food solution into a clean test tube.
2. Add ***2 cm^3*** of **Sodium Hydroxide (NaOH) solution** and mix thoroughly.
3. Add ***2–3 drops*** of **1% Copper(II) Sulphate ($CuSO_4$) solution** down the side of the test tube.
4. Shake the mixture gently and observe for color changes.

Observation (Positive)	Observation (Negative)	Deduction / Conclusion
Color changes from blue to **violet / purple / lilac**.	Solution remains **blue**.	Protein present (Positive) / Protein absent (Negative).

3. Vitamin C (Ascorbic Acid)

BIOLOGICAL IMPORTANCE

Acts as a vital antioxidant and co-factor for collagen synthesis (connective tissue integrity), promotes rapid wound healing, and strengthens overall immunity.

SIGNS, SYMPTOMS & DEFICIENCY

Deficiency: Scurvy.
Symptoms: Bleeding/swollen gums, delayed wound healing, internal hemorrhaging (subcutaneous spot bleeding), joint pain, and anemia.

Summarised Practical Procedure (DCPIP Test)

1. Pipette ***1 cm^3*** of **0.1% DCPIP solution** (blue dye) into a clean test tube.
2. Add the liquid test sample (e.g., fresh fruit juice) **drop by drop** using a calibrated dropper, gently shaking after each drop.
3. Count and record the minimum number of drops required to completely decolorize the DCPIP.

Observation (Positive)	Observation (Negative)	Deduction / Conclusion
The blue DCPIP dye is **decolorized** (turns colorless or light pink).	DCPIP solution remains **blue** despite adding excess drops.	Vitamin C present (Positive) / Vitamin C absent (Negative).

4. Reducing Sugars (e.g., Glucose, Fructose, Maltose, Lactose)

BIOLOGICAL IMPORTANCE

Serves as an immediate, readily accessible substrate for cellular respiration and metabolic energy release across tissues.

SIGNS, SYMPTOMS & DEFICIENCY

Deficiency: Hypoglycemia / General Energy Deprivation.
Symptoms: Sudden fatigue, muscle weakness, dizziness, sweating, trembling, and fainting.

Summarised Practical Procedure (Benedict's Test)

1. Add ***2 cm^3*** of food solution into a clean test tube.
2. Add ***2 cm^3*** of **Benedict's reagent**.
3. Place the test tube in a **boiling water bath** (***100°C***) for 3–5 minutes.

Observation (Positive)	Observation (Negative)	Deduction / Conclusion
Blue → Green → Yellow → Orange → **Brick-red precipitate**.	Solution remains **clear blue**.	Reducing sugar present (Positive) / Absent (Negative).

5. Non-Reducing Sugars (e.g., Sucrose)

BIOLOGICAL IMPORTANCE

Acts as the primary form of carbohydrate transport in plant phloem tissues and serves as a major energy storage molecule in plants.

SIGNS, SYMPTOMS & DEFICIENCY

Deficiency: Hypoglycemia / Energy Deprivation.
Symptoms: Lethargy, weakness, weight loss, and reduced metabolic activity.

Summarised Practical Procedure

1. Confirm negative result for reducing sugars first (solution remains blue upon heating with Benedict's).
2. Add ***2 cm³*** of fresh food sample + ***1 cm³*** of **dilute Hydrochloric acid (HCl)** into a test tube and heat in a boiling water bath for 2–3 minutes (*hydrolysis stage*).
3. Allow to cool, then add pinch of **Sodium Hydrogen Carbonate ($NaHCO_3$)** or dilute NaOH until effervescence ceases (*neutralization stage*).
4. Add ***2 cm³*** of **Benedict's reagent** and boil in a water bath for 3–5 minutes.

Observation (Positive)	Observation (Negative)	Deduction / Conclusion
Formation of a **brick-red precipitate** after hydrolysis.	Solution remains **blue** after hydrolysis & testing.	Non-reducing sugar present (Positive) / Absent (Negative).

Quick Reference Summary Table

Food Nutrient	Reagents Used	Heat Needed?	Positive Result Key Term	Associated Deficiency
Starch	Iodine Solution	No	**Blue-black** color	Marasmus
Proteins	NaOH + 1% $CuSO_4$ (Biuret)	No	**Violet / Purple / Lilac**	Kwashiorkor
Vitamin C	0.1% DCPIP Solution	No	**Decolorized** (Blue → Colorless)	Scurvy
Reducing Sugars	Benedict's Reagent	**Yes** (Water Bath)	**Brick-red precipitate**	Hypoglycemia
Non-Reducing Sugars	Dil. HCl, $NaHCO_3$, Benedict's	**Yes** (Hydrolysis + Bath)	**Brick-red precipitate**	Hypoglycemia

Possible UNEB Practical Scenarios & Key Examination Pitfalls

1. Common Candidate Errors Penalized by Examiners

- **Direct Heating:** UNEB penalizes candidates who heat test tubes containing Benedict's reagent directly over a naked Bunsen flame. Always specify heating in a *boiling water bath*.
- **Omission of Neutralization:** Failing to neutralize the acid after hydrolysis in non-reducing sugar tests will yield false negative results because Benedict's reagent cannot work in acidic conditions.
- **Vague Color Terminology:** Loose terms like "black", "dark blue", or "red solution" lead to loss of marks. Use precise terms: **"Blue-black"** for starch, **"Brick-red precipitate"** for reducing sugars, and **"Violet/Purple"** for proteins.

2. Typical UNEB Experimental Scenarios

- **Comparative Vitamin C Content:** Comparing drops of different fruit juices required to decolorize DCPIP. (*Rule: Fewer drops needed = higher concentration of Vitamin C*).
- **Unknown Solution Identification:** Providing unknown solutions labelled A, B, C and requiring candidates to systematically identify present nutrients using a sequence of tests.
- **Enzymatic Hydrolysis vs Acid Hydrolysis:** Testing starch breakdown after mixing with amylase/saliva over time using Benedict's and Iodine tests simultaneously.

Experimental Control Variables (Maintained Constant)

To ensure valid, reliable, and fair comparative testing across different food samples, the following control variables must be strictly held constant:

- **Volume and Concentration of Food Samples:** Using an exact, measured volume (e.g., ***2 cm^3***) or uniform mass of food extracts.
- **Reagent Quantities:** Standardizing the drops or volumes of reagents added (e.g., exactly ***2 cm^3*** of Benedict's or exactly ***1 cm^3*** of DCPIP).
- **Water Bath Temperature & Heating Duration:** Maintaining the boiling water bath at ***100°C*** and keeping heating times identical (e.g., exactly 3–5 minutes).
- **Completeness of Neutralization:** Adding precise quantities of ***$NaHCO_3$*** until effervescence completely stops to maintain identical pH conditions.
- **Dropper Calibration / Drop Size:** Utilizing identical droppers to maintain consistent drop volumes when measuring Vitamin C against DCPIP.